KOHSHIN

# HS-PHB series

RoHS compliant

**Small current range**
**PCB-mounting type**

## HS-PHB



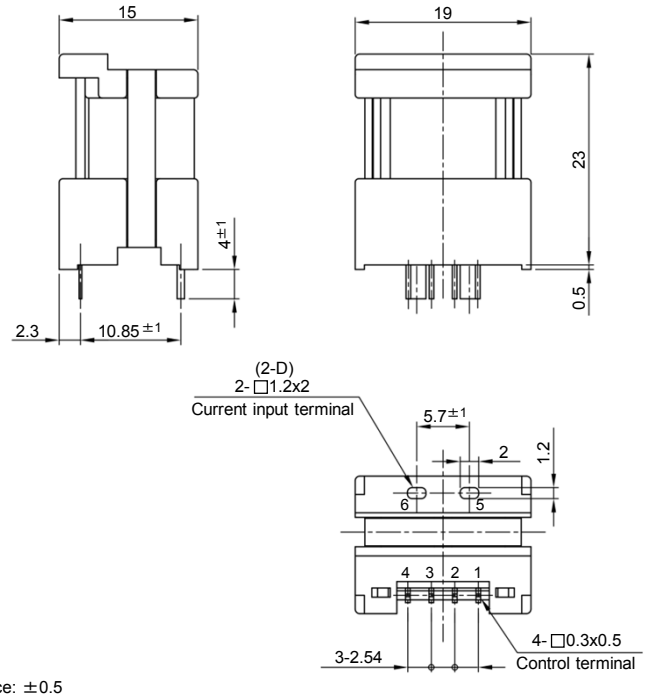
- Rated current 35A ~ 50A
- Realized high precision and compact size
- Superior in response, linearity and temperature characteristics

Applications

Inverters, Servo drivers, Power supply equipment, NC machine tools

## Dimensions

(mm)




Dimensions of Current Input Terminals

| Size of primary winding | Width D |
|---|---|
| Φ1.3 | Φ1.3 |
| □1.2 x 2 | □1.2 x 2 |

Terminal No.
1 - (-) terminal
2 - GND
3 - (+) terminal
4 - Output
5 - (+) input
6 - (-) input

Weight : 12g

General tolerance: ±0.5

## Specification

Ta=25℃

| | Voltage output type | | |
|---|---|---|---|
| Type | HS-PHB35V4B15 | HS-PHB40V4B15 | HS-PHB50V4B15 |
| Rated current [ If ] | ±35A | ±40A | ±50A |
| Continuously flowing DC current | ±25.2A | ±28.8A | ±36A |
| Saturation current [ Is ] | ±87.5A | ±100A | ±125A |
| Linearity limits | 0~±70A | 0~±80A | 0~±100A |
| Size of primary winding | Φ1.3 | □1.2 x 2 | □1.2 x 2 |
| Turns | 1 | 1 | 1 |
| Rated output [ Vh ] | ±4V±1.5% (RL=10kΩ) | | |
| Residual output [ Vo ] | Within ±30mV | | |
| Output linearity | Within ±0.5% | | |
| Response time | Within 3μs (at di/dt=If/μs) | | |
| Response performance | Within 20% | | |
| Hysteresis voltage range | Within 50mV | | |
| Output Temp. Coef. | Within ±0.04%/℃ | | |
| Residual output Temp. Coef. | Within ±1mV/℃ | | |
| Control power supply | ±15V±5% | | |
| Consumption current | 20mA+(Input current x N)/1270 | | |
| Operating Temp. | -10℃~+80℃ | | |
| Storage Temp. | -15℃~+85℃ | | |
| Dielectric withstand voltage | 2500V AC 50/60Hz 1minute | | |
| Insulation resistance | Not less than 500MΩ 500V DC | | |

Note1) The indicated residual output is the one after the core hysteresis is removed.

Note2) Energization time of saturation current shall be within 1 second.

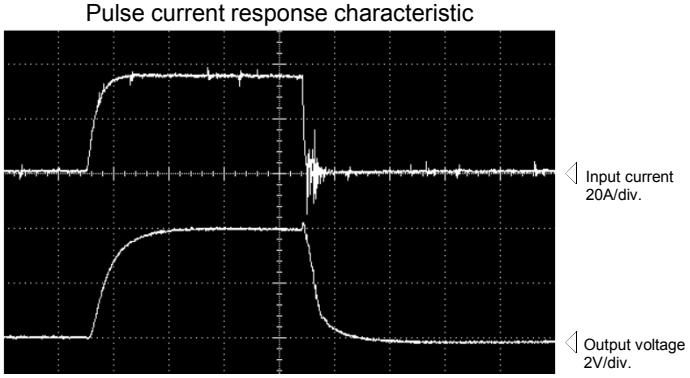
Note3) Energization time of continuous live DC current x150% shall be within 1 minute.

## Characteristics chart

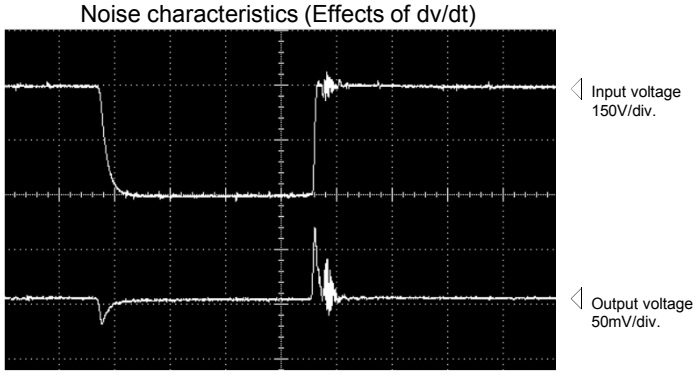
HS-PHB35V4B15 (RL=10kΩ)

Time base: 5μs/div.

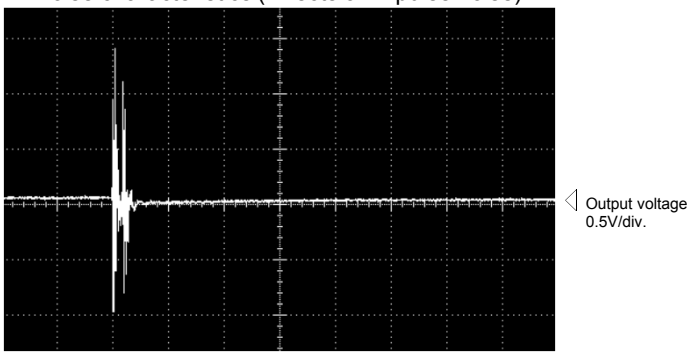
Pulse current response characteristic




Noise characteristics (Effects of dv/dt)

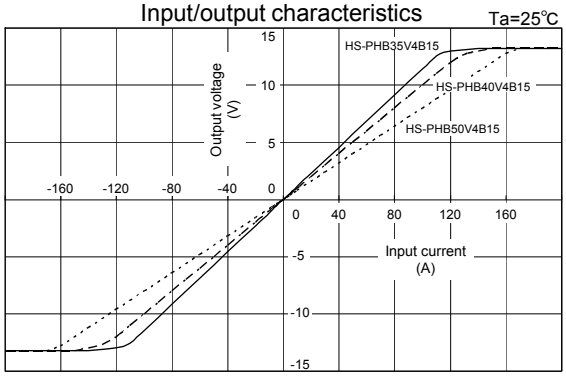



Noise characteristics (Effects of impulse noise)




Input/output characteristics

Ta=25℃




Note: The marks " ◁ " means 0V or 0A.